

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

Shanika (192511387)

I Betty A. (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Betty

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Smart Home Automation System

A smart home automation system controls lighting, temperature and security using sensors and AI

Types of Intelligent Agent:

i) Simple Reflex Agent:

- * It acts only based on the current input
- * It uses if-then rules
- * It does not remember previous events

Eg: motion is detected, then light will turn on.

motion is not detected, then light turn off.

Advantages:

- * Simple and fast
- * low cost

Disadvantages:

- * cannot learn uses habits
- * cannot handle complex situation

ii) Model - Based Reflex Agents

- * maintain an internal model of the environment

Eg: Remember that a room was occupied recently

Advantages:

Better decision making

Disadvantages:

more complex than a simple reflex agent.

iii) Goal Based Agent:

makes decision to achieve a specific goal

Eg: maintains a comfortable temperature and ensures home security.

* can plan before taking actions

iv) Utility - Based Agent :

* select the action that provide the highest overall benefit

* Balance multiple objectives

Eg: The agent decides whether to

- Turn on the AC for comfort
- Reduce AC usage to save electricity

Advantages:

make intelligent trade-off

Disadvantages:

Utility calculation can be complex:

Best Approach:

A hybrid of model-based and utility based agent is the most effective because it remembers previous information and preferences.

Q. A Robot placed in an unknown environment must search for the exit without knowing the layout. AI search algorithm helps the robot explore the environment and find the correct path.

Uniform cost search (UCS):

Uniform cost search expands the node with lowest cumulative path cost.

Working:

- * Start from the initial position
- * calculate the total cost of each position path.
- * Always chooses the path with the minimum cost
- * continues until the exit is reached.

Eg: If one path cost 5 unit and another cost 8 unit, UCS selects the path with cost 5.

Advantage:

- * complete
- * finds the optimal path
- * suitable path cost vary

Disadvantages:

- * High memory usage
- * can be slow for large mazes.

Iterative Deepening Search (IDS)

Iterative Deepening search performs Depth first search with increasing depth limits.

Working:

Search depth = 0

if exist not found $\rightarrow$ depth = 1

Then depth = 2

continues until the exist found

Advantages:

- * Low memory usage
- * complete
- * suitable when solution depth is unknown

Disadvantages:

- * Repeat node expansion
- * more computation time than BF

time and space complexity

UCS:

Time complexity - $O(b^n(1+c^*/\epsilon))$

Space complexity - $O(b^n(1+c^*/\epsilon))$

ID3:

Time complexity - $O(b^nd)$

Space complexity - $O(bd)$

Where,

b = branching factor

d = depth of solution

c^* = optimal path cost

ϵ = minimum step cost

Types of Intelligent Agent

i) Simple Reflex Agent

- * It acts only based on the current input

* It does not remember previous events.

ii) Model - Based reflex Agent

* maintain an internal model of the environment

iii) Goal Based Agent:

* makes decision to achieve a specific goal.

iv) Utility Based Agent:

* select the action that provides the highest overall benefit

* Balance multiple objectives.

Best combination:

i) Goal Based + uniform cost search:

when path cost vary

Eg: different energy cost

because it finds the lowest cost route.

ii) Goal Based + Iterative Deepening search:

When all moves have equal cost
and memory is limited because it
was much less space.